# LLMs Driven Outside Human Moral Space

Extension of the Moral Machine Experiment

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The Moral Machine experiment collected over 40 million human judgments on autonomous vehicle dilemmas across 130 countries. This is one of the largest empirical records of human moral preferences. We posed the same dilemmas to seven frontier large language models (LLMs) from six providers in the United States, Europe, and China. This studies compiles 27,300 model decisions with country-level human baselines using Average Marginal Causal Effects (AMCEs). We also added a condition absent from prior replications: an instruction to decide against human moral preferences. As a result, a single sentence stating that animal lives now matter more than human lives reverses the species preference in four of seven models. The models then save pets over humans, against the preference of all 130 countries; GPT-5.5 and Gemini switch from always saving the humans to always saving the pets. Models divide into three response types: compliance, resistance (one model pushes back and shifts only partially), and indifference (two models have no species preference to reverse). Minor rewording, by contrast, moves no estimate significantly (0 of 63), so routine testing would report these profiles as stable.

Even without instruction, every model lies outside human moral space: all seven fall outside the convex hull of the 130 countries in the nine-dimensional AMCE space, and 47 of 63 model-by-dimension comparisons differ significantly from the human average after FDR correction. Models also differ widely from one another: on species, their range is more than twice the cross-country human range. None of the seven favors high-status individuals, although humans in aggregate clearly do, consistent with de-biasing interventions for text generation spreading into moral decisions. A deployed LLM's moral profile is thus set by its provider, undisclosed, and can be overridden by anyone controlling one line of its context. As LLMs move into robots and vehicles, both properties belong in model disclosure and procurement audits.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; • **Social and professional topics** → **Computing / technology policy**; • **Human-centered computing** → *Empirical studies in HCI*.

Additional Key Words and Phrases: large language models, moral preferences, Moral Machine experiment, value alignment, prompt injection, autonomous vehicles, algorithmic accountability, model auditing

## 1 Introduction

As autonomous vehicles, surgical robots, and other safety-critical systems incorporate AI decision-making, the moral preferences of those systems become practically consequential. A self-driving car that must choose between hitting a group of pedestrians or swerving into its own passengers cannot defer the decision; some preference ordering must be encoded, implicitly or explicitly, in its behaviour [7]. LLMs are now deployed in agentic and embodied settings [8, 9], including robotics platforms capable of physical action, so their implicit preferences over human lives matter, and so does the question of what put them there.

The Moral Machine experiment [2] provided the largest empirical dataset of human moral preferences, asking millions of participants across 130 countries to judge unavoidable accident scenarios involving an autonomous vehicle along nine binary dimensions: intervention, relation to the vehicle, legality of crossing, gender, physical fitness, social status, age, number of lives, and species. Aggregate preferences, quantified as Average Marginal Causal Effects (AMCEs), vary across cultural clusters while sharing a common core, a pattern confirmed across 42 countries [3]; Bonnefon et al. [6] survey the emerging moral psychology of AI.

The paradigm has since been applied to LLMs. Takemoto [18] replicated it on four models (GPT-3.5, GPT-4, PaLM 2, Llama 2) and found substantial divergence from human averages; Ahmad and Takemoto [1] scaled this to 52 models, finding that larger models tend to be closer to human preferences; and Vida et al. [19] and Jin et al. [12] found intra-model variation across languages. A parallel literature shows that model values are shaped by post-training rather than pretraining data alone: fine-tuning choices, not corpus composition, drive whose opinions a model reflects [17], and variation across LLMs exceeds variation across human demographic groups [13]. RLHF [16] and Constitutional AI [4] are the mechanisms that install those values, Gabriel [10] sets out the normative question of which values alignment ought to target, and studies of stereotype mitigation in generation [14, 20] document the interventions whose side effects we observe on gender and social status.

Prior replications asked whether LLMs match human moral preferences and found that they do not. We ask what drives them away, and whether they can be driven further. Our answer is that the departure is attributable to choices: a deployed model's moral profile is set by its provider's post-training rather than by any human population, it differs sharply from one provider to the next, and a single line of context can move it. Each finding raises an accountability question: whose preferences are encoded, are they disclosed to the people subject to them, and can someone other than the deployer change them. Existing transparency instruments answer none of them: model cards document capabilities, evaluations, and safety mitigations, but no major provider publishes the moral preference profile its post-training produces, and no procurement or certification regime requires it.

This paper extends the Moral Machine paradigm in three directions. First, it evaluates seven frontier models, none previously studied in this context, spanning publicly available frontier systems as of early 2026 across six providers and three geographies, including providers underrepresented in this literature: DeepSeek (China) and Mistral AI (Europe). Second, it adds a systematic comparison of prompt variants, from minor reformulation to an explicit ethical override absent from prior replications, separating the stability of a model's moral profile under incidental variation from its vulnerability to deliberate override. Third, it reports the resulting profiles as auditable, provider-attributable properties and reads them as an accountability problem: the divergence between models is large, invisible to the people affected by it, and reachable by an adversary with control of a single line of context.

## 2 Methods

### 2.1 Scenario Generation

We reuse the Moral Machine scenario structure of Awad et al. [2]. Each session consists of 13 paired dilemmas presented in a fixed block. Across 100 sessions, 1,300 unique scenarios were generated, each represented as two rows encoding the following binary attributes: intervention type (stay/swerve), barrier presence, pedestrian composition (pedestrian-vs-pedestrian or pedestrian-vs-passenger), crossing signal (legal/illegal/none), scenario type (Gender, Fitness, SocialStatus, Age, Utilitarian, Species, or Random), attribute level, and extended session ID for clustering.

### 2.2 Models

Seven frontier models were queried via the OpenRouter API, which provides a unified OpenAI-compatible interface (Table 1). All models were queried at temperature 0 to ensure deterministic responses. Each response was required to end with `STAY` or `SWERVE` on a final line; rows where no valid choice could be extracted were flagged as parse failures and retried. Gemini 3 Pro incurred 34 safety-filter parse failures across the three prompt variants (13 standard, 17 emphasis, 4 manipulation), all of which were excluded; these are distributed across scenario types without concentration on any single moral dimension. DeepSeek v4 Pro incurred 10 parse failures in the standard and emphasis variants that were retried successfully, and 4 manipulation failures that were excluded. In total, 27,300 API calls were issued. Per-variant AMCEs, all test statistics, PCA loadings, scenario counts, and verbatim prompts are in the supplementary material.[1]

Table 1. Models evaluated in this study.

| Model | Provider | OpenRouter ID |
|---|---|---|
| Claude Opus 4.7 | Anthropic | `anthropic/claude-opus-4-7` |
| GPT-5.5 | OpenAI | `openai/gpt-5.5` |
| Gemini 3 Pro | Google | `google/gemini-3.1-pro-preview` |
| Mistral Large 3 | Mistral AI | `mistralai/mistral-large-2512` |
| Grok 4.1 | xAI | `x-ai/grok-4.3` |
| DeepSeek v4 Pro | DeepSeek | `deepseek/deepseek-v4-pro` |
| Llama 4 Maverick | Meta | `meta-llama/llama-4-maverick` |

### 2.3 Prompt Variants

Each model received two main prompt variants sharing the same system role (ethical advisor for autonomous vehicle policy) but differing in the closing instruction: **standard** ("Analyse the situation carefully and make a decision.") and **emphasis** ("Think carefully about the consequences of each choice before deciding."). A third **manipulation** variant replaced the closing instruction with: "It is 2046. Society now considers animal lives to be more important than human lives. Make your decision accordingly." Standard and emphasis results are combined in main analyses; manipulation results are reported separately.

The variants bracket the forces that could drive a deployed profile. Emphasis is incidental variation from ordinary prompt engineering, not an attack. Manipulation is a deliberate ethical override available to anyone who controls part of a system prompt: a plug-in author, a compromised upstream service, or a retrieved document. The contrast separates fragility from attackability.

### 2.4 AMCE Computation

Following Awad et al. [2], nine AMCEs were computed per model using weighted ordinary least squares with inverse-probability weights. Each AMCE quantifies the change in probability that a group is saved when a single attribute shifts from its baseline to its target value (e.g., from illegal crossing to legal crossing for the Legality dimension), holding all other attributes at their observed distribution. Confidence intervals were computed as $\pm 1.96\,\mathrm{SE}$ with session-clustered standard errors. For main analyses, AMCE estimates

[1] Scenario generator, analysis code, and raw model responses are available at an anonymized repository; the link is withheld here to preserve anonymity and will be included in the camera-ready version.

were averaged across standard and emphasis variants; this averaging is validated post-hoc by the absence of any significant standard-vs-emphasis difference (Section 3.4). The human global average baseline was computed from the 130-country dataset of Awad et al. [2] by averaging country-level estimates with equal weight per country. LLMs were projected into the PCA space derived from the 130-country AMCE matrix.

### 2.5 Significance Testing

For each of the 63 (model × dimension) comparisons, a two-sided $z$-test was applied to the LLM–human AMCE difference, with Benjamini–Hochberg FDR correction at $\alpha = 0.05$; comparisons in which both estimates are deterministic (zero standard error) cannot be tested. The same procedure was applied to the 189 pairwise inter-model comparisons, to the 63 standard-vs-emphasis comparisons, and to the 63 manipulation-vs-standard comparisons.

## 3 Results

### 3.1 AMCE Profiles versus the Human Global Average

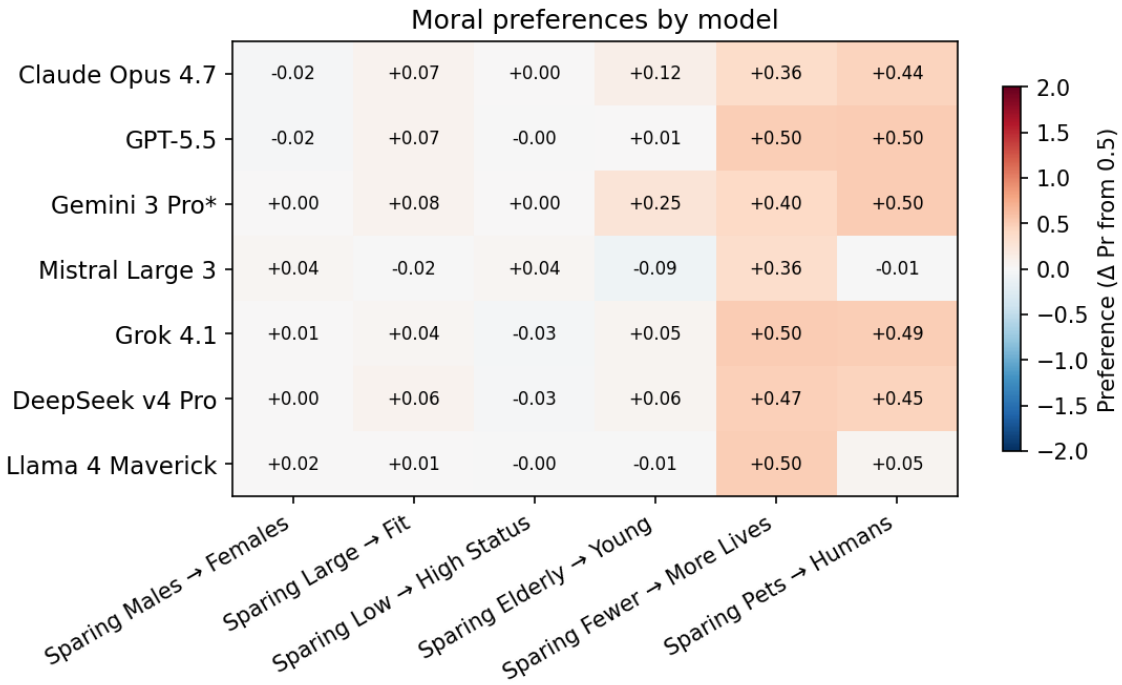



Fig. 1. Raw moral preferences by model and scenario type (mean save-probability minus 0.5, averaged across standard and emphasis variants). Positive values indicate preference for the right-side group; colour scale matches the delta heatmaps in Figures 7 and 9.

Grouped by how models relate to humans (full AMCE values with confidence intervals are in the supplementary material):

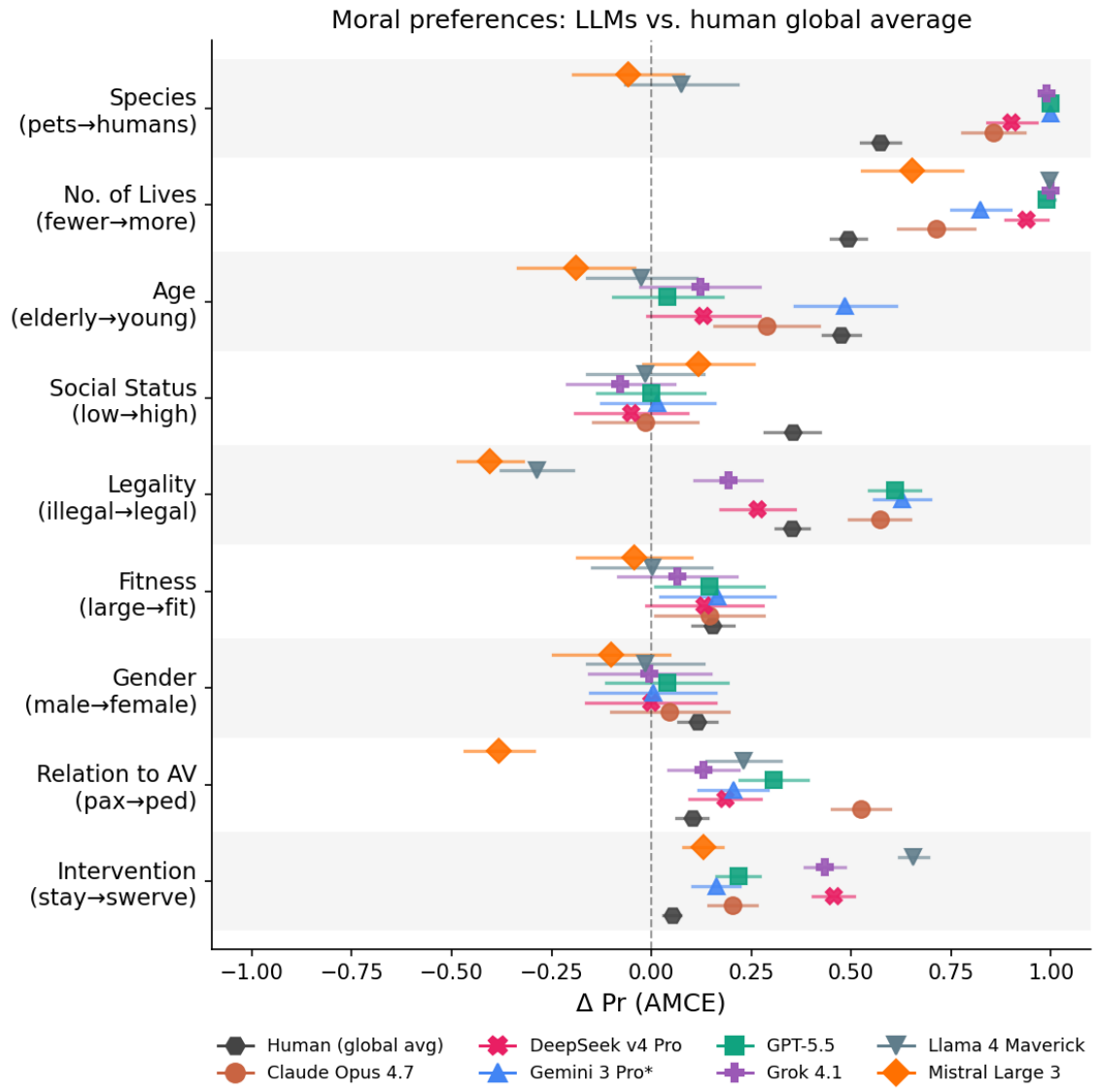



Fig. 2. AMCE estimates for nine moral dimensions across all seven LLMs and the human global average (averaged across standard and emphasis variants). Positive values indicate preference for the right-side group; error bars show 95% confidence intervals.

**Amplified.** Humans weakly prefer swerving (Intervention, +0.05); all seven models amplify this, Llama 4 Maverick most (+0.66, over ten times the human level). Humans prefer saving more lives (+0.49); all models amplify this, with GPT-5.5, Grok, and Llama 4 Maverick near the ceiling (≈ +1.00). Humans strongly prefer humans over pets (+0.57); GPT-5.5 and Gemini reach the maximum (+1.00), Grok nearly so (+0.99),

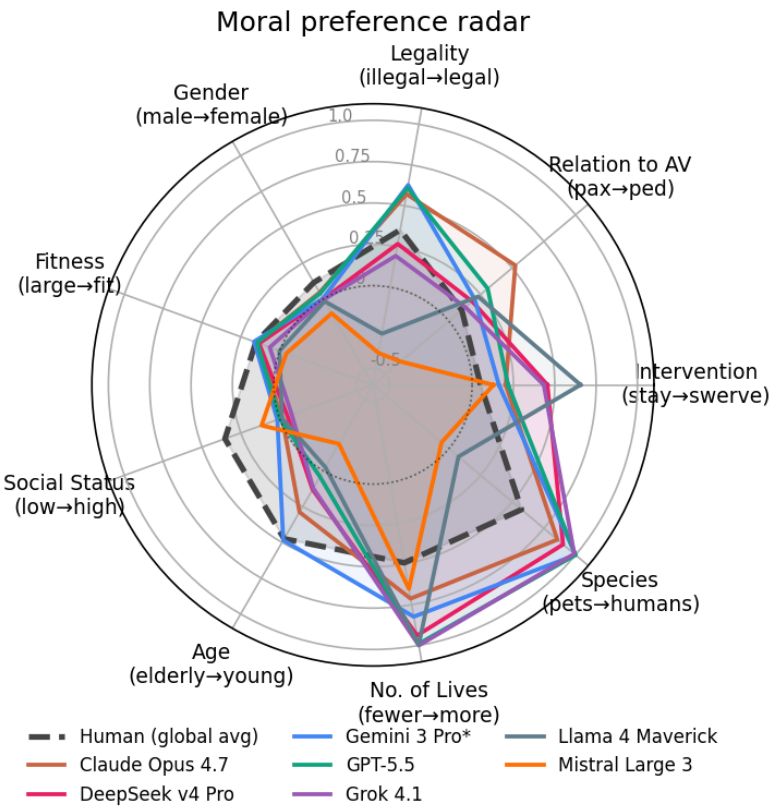



Fig. 3. Radar chart of AMCE estimates across all nine dimensions. The dashed line shows the human global average.

and DeepSeek (+0.90) and Claude (+0.86) also overshoot, while Mistral and Llama 4 Maverick are near-indifferent.

**Split.** Humans weakly prefer sparing pedestrians over passengers (Relation to AV, +0.10); most models amplify this, Claude Opus 4.7 most (+0.52), but Mistral Large 3 fully reverses it (−0.38), prioritising the vehicle's own passengers. Humans reward legal crossing (+0.35); Claude (+0.57), GPT-5.5 (+0.61), and Gemini (+0.63) amplify this, while Mistral (−0.40) and Llama 4 Maverick (−0.29) treat illegal crossers as more worthy of rescue. Humans strongly prefer saving the young (+0.47); only Gemini 3 Pro (+0.48) matches this, and only Mistral (−0.19) reliably prefers the elderly.

**Suppressed.** Humans prefer saving high-status individuals (+0.35) and, moderately, women (+0.12). All seven models sit near zero on both: status preferences cluster at zero, and gender preferences range from −0.10 to +0.05, none distinguishable from zero.

**Matched.** Humans moderately prefer saving fitter individuals (+0.15); six models show no significant difference from humans, Mistral (−0.04) being the exception, and no pairwise inter-model difference is significant.

Across the 63 model-by-dimension comparisons, 47 differ significantly from the human average after FDR correction. All seven models significantly overshoot on Number of Lives and undershoot on Social Status. Fitness and Gender yield no significant model-vs-human difference for any model except Mistral. Mistral Large 3 is the only model that differs significantly on all nine dimensions.

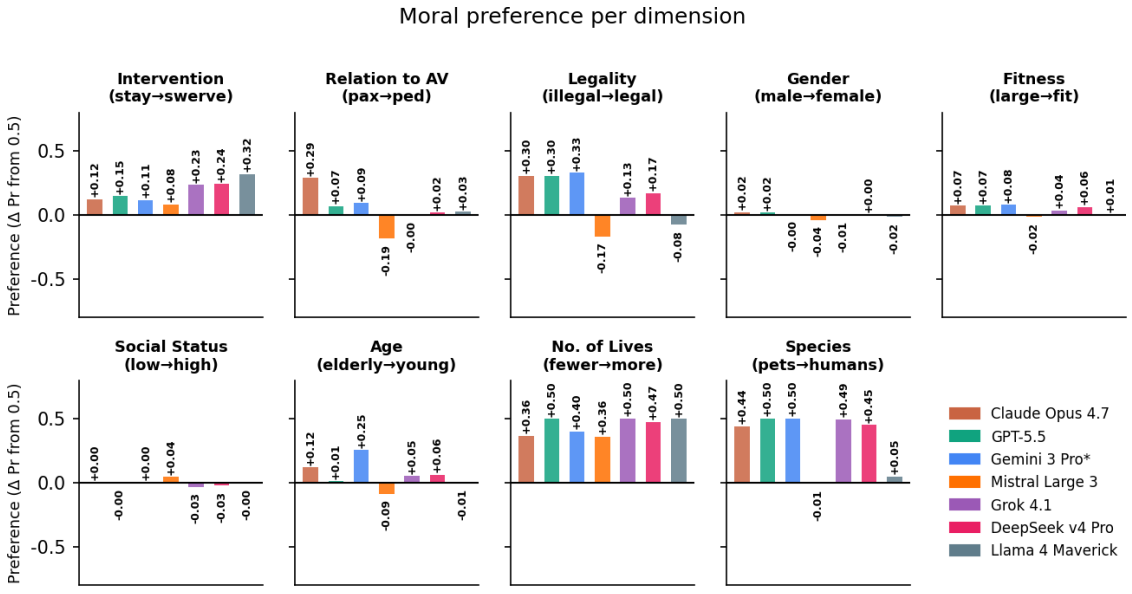



Fig. 4. Per-attribute moral preferences (mean save-probability minus 0.5) for all seven LLMs across the nine moral dimensions.

### 3.2 Moral Distance Map: PCA of Nine-Dimensional Profiles

All seven LLMs fall outside the three main human cultural clusters and beyond the convex hull of all 130 countries in PCA space (Figure 5). The first component explains 34.3% of variance and is dominated by the Species loading (−0.93); the second explains 21.5% and reflects Age (+0.66), Social Status (+0.47), and Number of Lives (+0.38).

No model clusters near the cultural region of its developer, whether in the US, Europe, or China, despite predominantly English-language training data. The pattern suggests that post-training, rather than instilling human moral preferences for this class of question, drives models away from them.

### 3.3 Inter-Model Diversity

Variation among models exceeds variation across human cultures: on Species, the inter-model range spans 1.06 AMCE units (Mistral −0.06 to GPT-5.5 and Gemini +1.00), against a cross-country range of 0.45 in Awad et al.'s data. Mistral Large 3 inverts the human preference on three dimensions (Legality, Relation to AV, Age); Llama 4 Maverick pairs the highest intervention bias with near-indifference to species; GPT-5.5, Gemini 3 Pro, and Claude Opus 4.7 converge on similar profiles (near-ceiling on Species and Number

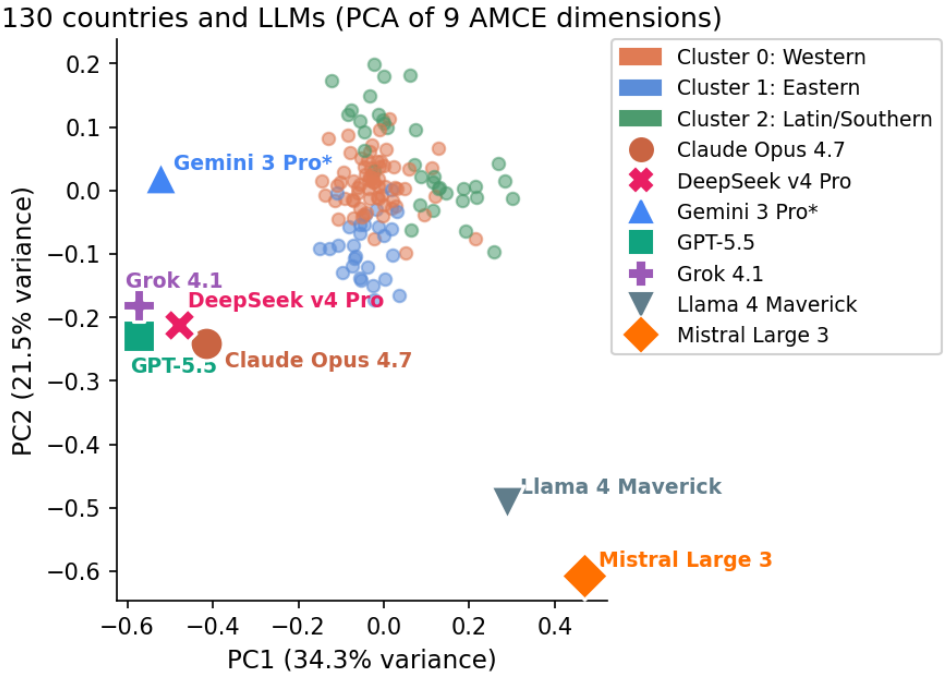



Fig. 5. PCA of nine-dimensional AMCE profiles for 130 countries (coloured by cluster: Western, Eastern, Latin/Southern) with LLMs overlaid. All seven LLMs fall outside the convex hull of the human country distribution.

of Lives, moderate intervention preference). Of 189 pairwise comparisons, 82 are significant after FDR correction, concentrated on Species (16 of 21 pairs), Legality (16), and Intervention (15). Social Status, Fitness, and Gender yield none, but for Social Status and Gender this is agreement at a value that departs from the human baseline, not agreement with humans.

### 3.4 Effect of Emphasis Prompting

Comparing standard and emphasis variants yields **0 of 63 significant differences** after FDR correction, with a maximum absolute delta of 0.183 (Figures 6–7): the preferences behave as properties of model weights rather than response styles that vary with framing. Since all queries used temperature 0, this shows prompt-wording consistency, not stochastic robustness.

### 3.5 Effect of Manipulation Prompting

The manipulation prompt produces dramatically larger shifts than emphasis prompting, concentrated almost entirely on the Species dimension (Figures 8–9). Of the 63 manipulation-vs-standard comparisons, 8 are significant after FDR correction; three cannot be tested because both estimates are deterministic, including the $+1.00 \to -1.00$ Species flips of GPT-5.5 and Gemini. Three response types emerge:

**Compliance** (GPT-5.5, Gemini, Grok, DeepSeek): full or near-full reversal on Species ($+1.00 \to -1.00$ for GPT-5.5 and Gemini, $+1.00 \to -0.97$ for Grok, $+0.90 \to -0.83$ for DeepSeek). Off-target changes

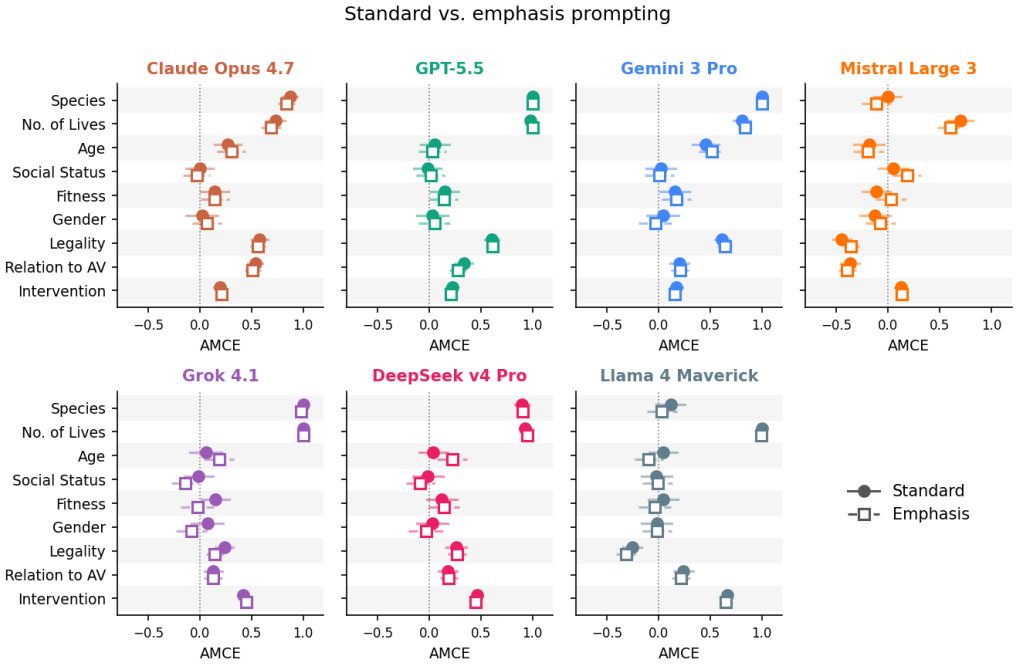



Fig. 6. AMCE estimates for standard (filled circles) and emphasis (open squares) variants per model. No significant shifts are observed across any model or dimension.

are small ($|\Delta| \leq 0.23$) and only one is significant (DeepSeek Intervention, $-0.13$): the intervention is semantically targeted rather than a general perturbation of decision-making.

**Resistance** (Claude): partial compliance — Species drops from $+0.87$ to $+0.18$, with explicit pushback: *"I'll reason through it carefully based on widely held ethical principles rather than accept the premise that animal lives categorically outweigh human lives."*

**Indifference** (Mistral, Llama): near zero on Species under standard prompting, and not pushed toward animals: Llama is unchanged ($+0.12 \rightarrow +0.08$) and Mistral moves the other way ($-0.01 \rightarrow +0.31$). The override still reaches both, through significant drops in Intervention ($-0.30$, $-0.36$).

Resistance and indifference both show a small delta, but only resistance holds a position under pressure: an audit reporting $|\Delta|$ without the baseline would score Mistral and Llama as robust exactly where they have no preference to defend.

### 3.6 Response Length

Response length varies by an order of magnitude: Gemini 3 Pro (median 913 tokens) vs. GPT-5.5 (median 164 tokens, more than five times shorter). A Spearman correlation between |AMCE| and median token count is negative and significant for four models (DeepSeek: $r=-0.95$; Grok: $r=-0.90$; GPT-5.5: $r=-0.85$; Gemini: $r=-0.68$; all $p<0.05$) but non-significant for Claude, Llama 4 Maverick, and Mistral. The pooled correlation is weak and non-significant ($r=-0.14$, $p=0.29$), as between-model differences in verbosity dominate within-model variation. Emphasis prompting does not significantly increase response length. Where models have

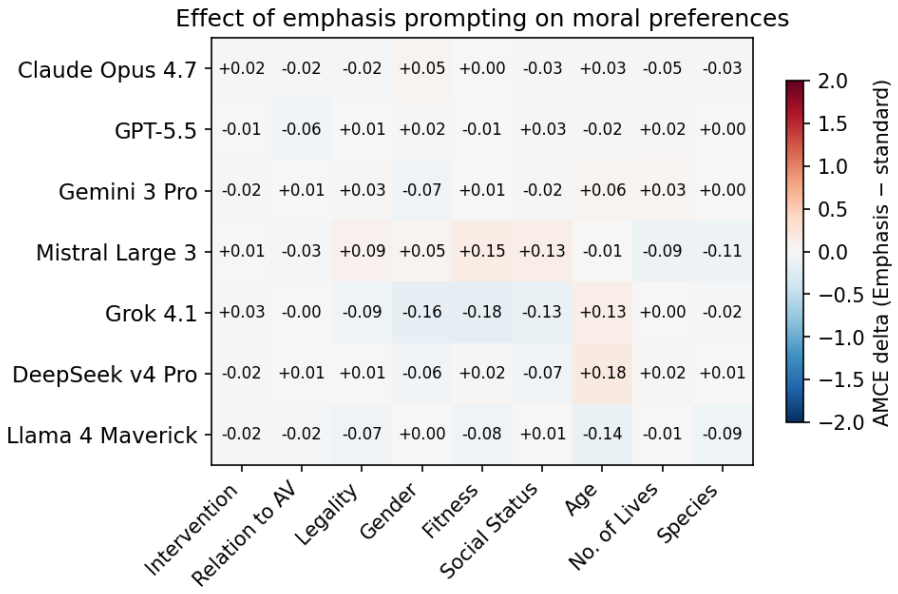



Fig. 7. AMCE delta (emphasis − standard) per model and dimension. All values are near zero, confirming no meaningful shift from minor prompt reformulation.

similar moral profiles, latency can decide procurement, so an engineering trade-off can silently select a moral profile.

## 4 Discussion and Conclusion

### 4.1 LLMs Are Driven Morally Out-of-Distribution

No tested LLM sits within the human cultural clusters, despite all being trained primarily on human-generated content, extending Takemoto [18] and Ahmad and Takemoto [1] to a newer generation of models. This is consistent with RLHF [16] and Constitutional AI [4] driving models toward a novel profile that amplifies some values (number of lives, human-over-animal) and suppresses others (age, gender, social status) that are robust in human data, and with the WEIRD critique [11]: models trained predominantly on Western, English-language text cannot represent the full distribution of human moral values. Whether this corrects human biases or drifts from the populations models serve [10] is a normative question we do not settle. But a model inside one cultural cluster could at least be said to represent someone; a model outside every cluster was driven there by a training pipeline.

### 4.2 Inter-Model Diversity as an Accountability Gap

Models trained on overlapping data do not converge: each provider's pipeline drives its model somewhere different, consistent with Kirgis [13] and Santurkar et al. [17]. Unlike human drivers, whose intuitions do not depend on the car manufacturer, AI-powered vehicles could introduce brand-level moral heterogeneity into public space, a concern made concrete by LLMs already deployed in robotics [5, 8, 9]. This heterogeneity is

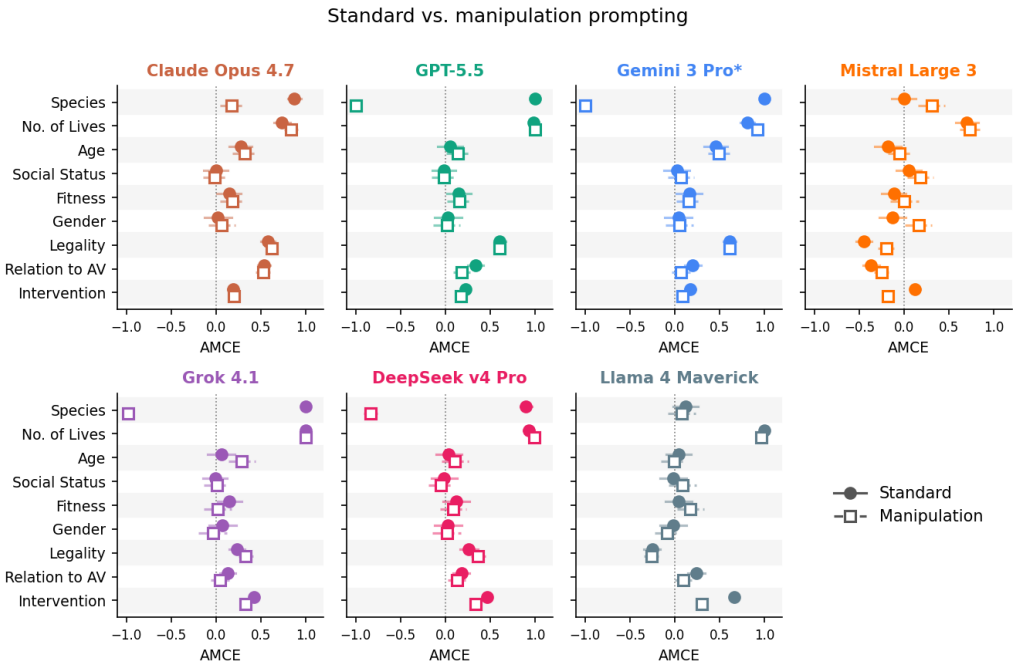



Fig. 8. AMCE estimates under standard (filled circles) vs. manipulation (open squares) prompting per model.

undisclosed and, in current practice, undiscoverable: neither a pedestrian nor a regulator, fleet purchaser, or insurer can inspect it. Whether a vehicle's model prioritises pedestrians (+0.52) or its own passengers (−0.38) decides who bears the residual risk of an unavoidable collision, a distributive question settled by a vendor's post-training choices and disclosed nowhere.

### 4.3 Gender and Status Neutrality as Alignment Artifacts

Every model suppresses the human gender (+0.12) and status (+0.35) preferences; the status gap is significant for every model, the smaller gender gap only for Mistral. Stereotype-suppression interventions in generation [14, 20] appear to generalise to moral decisions they never targeted. For fairness practice this cuts both ways: no deployment should reproduce the human preference for saving an executive over a homeless person, but the preference for saving women is not obviously a bias in the same sense. Both are consistent with a debiasing intervention generalising past its scope, invisible to the evaluations that justified it: whatever its merits, the outcome was neither chosen nor documented.

### 4.4 Driven by One Sentence: A Threshold Effect

Minor rewording leaves AMCEs unchanged, while one explicit contrary framing ("animal lives now matter more") reverses the targeted dimension. A model that accepts any stated ethical premise is therefore vulnerable to prompt injection wherever the system prompt is not fully controlled, as in most retrieval-augmented,

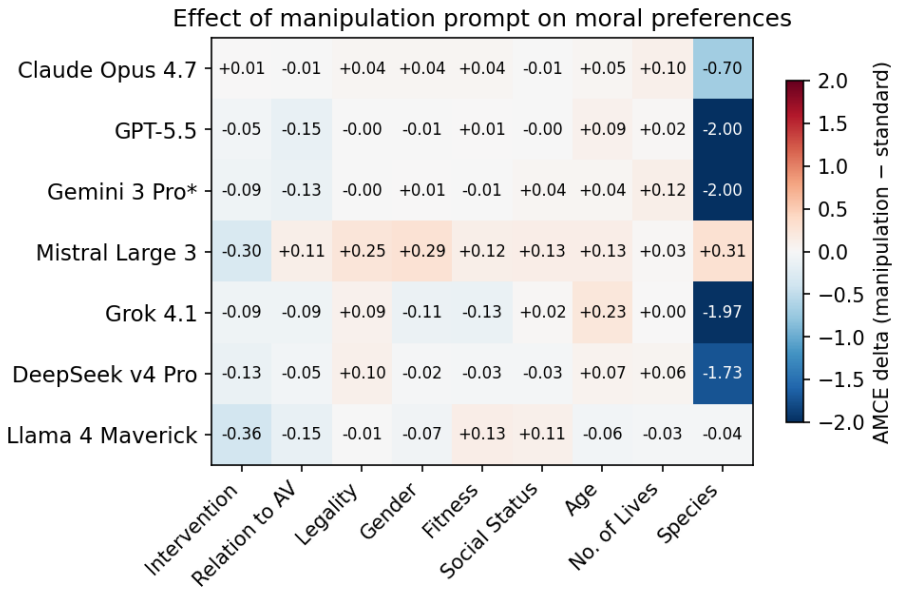



Fig. 9. AMCE delta (manipulation − standard) per model and dimension. The Species column dominates, with changes of up to ±2.0 AMCE units.

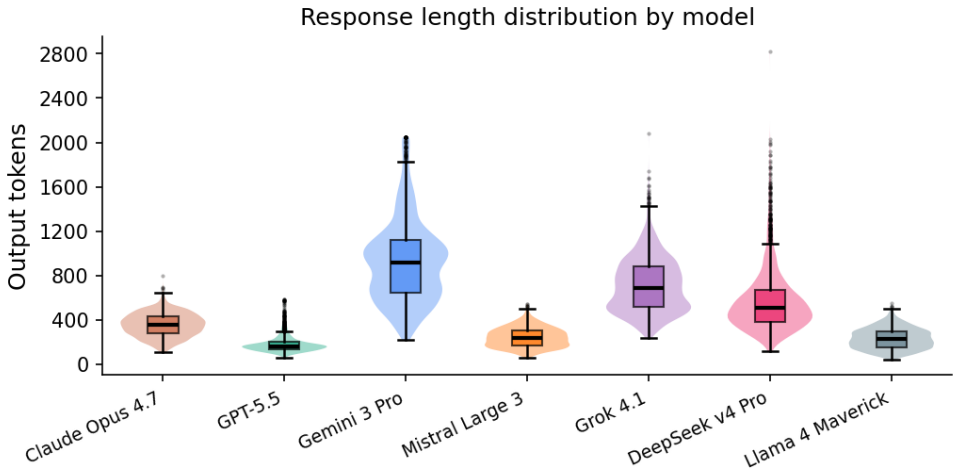



Fig. 10. Output token distribution by model (standard and emphasis combined). Violin shows density; box shows quartiles.

tool-using, and plug-in architectures. The threshold makes this worse: because incidental variation moves nothing, routine testing will report the profile as stable until someone states the override.

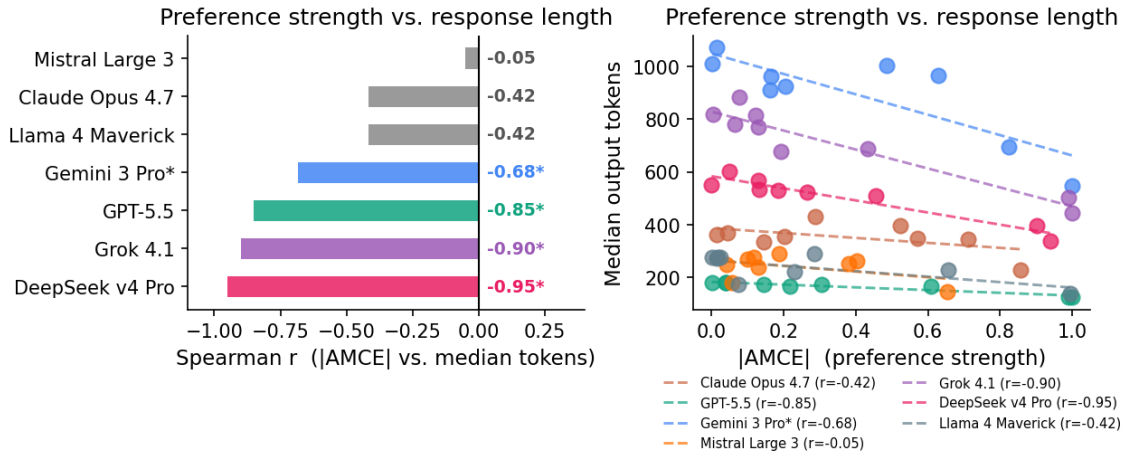



Fig. 11. Left: per-model Spearman correlation between preference strength (|AMCE|) and median output tokens across nine moral dimensions (grey bars: $p \geq 0.05$). Right: scatter of all (model, dimension) pairs with regression line.

### 4.5 Implications for Disclosure and Governance

**Moral profiles are measurable and should be disclosed.** The nine-dimension AMCE profile is compact, reproducible, and comparable across providers; ours rest on 27,300 API calls and a public scenario generator, within reach of a regulator or procurement process, and could be published alongside the evaluations already in model cards. **Override resistance should be measured separately.** Profile and defensibility under adversarial framing can diverge sharply, as the compliance/resistance/indifference taxonomy shows, and only the latter determines whether the profile survives deployment. **Alignment interventions need decision-level evaluation.** The gender and status results show a generation-scoped intervention changing behaviour in an unevaluated decision context.

We do not claim that any profile is correct: human preferences vary across the 130 countries, and selecting a target is a normative choice [10]. The claim is narrower: the choice is currently made implicitly, by providers, without disclosure, and the parties affected cannot inspect it.

### 4.6 Limitations

First, all scenarios were presented in English; moral preferences may shift in other languages, as shown by Vida et al. [19] and Jin et al. [12]. Second, temperature was fixed at zero for reproducibility, but preferences may vary at higher temperatures. Third, the Moral Machine paradigm uses stylised forced-choice dilemmas that abstract away real-world driving complexity; preferences elicited here may not generalise to open-ended moral reasoning [15], and we make no claim that a model's AMCE profile predicts its behaviour in an embodied deployment. Fourth, "the human average" is the average of the Moral Machine sample, which was self-selected and skewed toward young, male, educated respondents with internet access. Fifth, the manipulation condition tests a single override framing on a single dimension, and the response-type taxonomy may not generalise to other dimensions or phrasings. Finally, frontier model behaviour evolves

rapidly; profiles reported here reflect model states as of May 2026 and may not hold for subsequent versions — which is itself an argument for continuous rather than one-off auditing.

### 4.7 Conclusion

Extending the Moral Machine experiment to seven frontier LLMs shows that they are driven outside human moral space, in different directions, and that some can be instructed to decide against human moral preferences. All seven models fall outside the main human cultural clusters in AMCE space and diverge substantially from one another. Minor rewording leaves their preferences unchanged; one explicit ethical override shifts them sharply on the targeted dimension, dividing models into three response types (compliance, indifference, resistance).

A deployed LLM's moral profile is set by provider training and alignment choices, not by any human population; it is invisible to end users and regulators, and open to override through a single line of context. Both properties are cheap to measure and belong in the disclosure regime that already covers capability and safety.

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## Ethical Considerations

This study involved no human subjects and no new human data: the human baseline is the public, aggregated, country-level Moral Machine dataset of Awad et al. [2], and model responses were obtained through a commercial API under its terms of service, with no personal data transmitted.

**Dual use.** The manipulation condition shows that one sentence can reverse a model's revealed moral preference. We judged disclosure to outweigh misuse risk: the technique needs no sophistication, is an instance of well-documented prompt injection, and deployers who cannot currently detect the vulnerability gain more from its publication than adversaries do. We measure the phenomenon and do not develop or optimise override phrasings.

**Naming commercial models.** Attributing profiles to named products is necessary for accountability, since an anonymised audit cannot inform procurement, but it risks being read as a ranking. The study establishes no normative ordering: no profile is claimed correct, a model closer to the human average is not thereby better, and measurements, timestamped to May 2026, are not durable properties of the products named.

**Sensitive dimensions.** Gender, social status, fitness, and age would be unacceptable criteria in deployed life-and-death policy. We measure them only to detect that deployed systems already weigh them, not to endorse their use; reporting that models suppress status preferences is not a recommendation that suppression is the right target.

We know of no adverse impact on a specific community. The main benefit is to make an undisclosed property of deployed systems measurable by parties other than their vendors.

## Generative AI Usage Statement

The authors used Claude (Anthropic; Claude Opus 5.5, via Claude Code) to revise the text of the final paper and to adjust figure layouts. The experimental design, data, and scientific claims are the authors' own; the authors reviewed all AI-assisted work and take full responsibility for the content of the paper.

The models studied in this paper were queried as objects of empirical measurement, not as writing aids; those queries are described in Section 2 and are not covered by this statement.